Hexadecimal

Hexadecimal is also a number system. Humans can use hexadecimal, but computers cannot process it. Like denary numbers, hexadecimal numbers also need to be converted to binary to be processed by a computer.

Hexadecimal is a base-16 system and uses 16 symbols, these are the numbers 0–9 and the letters A–F. This is to allow the denary values 0–15 to be represented in hexadecimal. This is shown in Table 1.1.

Denary	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Table 1.1: Hexadecimal representations of denary numbers

The reason for using the symbols A–F for the denary numbers 10–15 is so only one symbol is needed for the denary number in hexadecimal.

SKILLS FOCUS 1.2

CONVERTING BETWEEN DENARY NUMBERS AND HEXADECIMAL

To convert a denary number to hexadecimal, it is easier to convert it to binary first. For example:

First convert the denary number 7 to 4-bit binary.

8	4	2	1
0	1	1	1

To convert the binary number to hexadecimal you can use simple addition again. 4 + 2 + 1 = 7. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 7 is also 7. Therefore, the denary number 7 is 0111 as binary and 7 as hexadecimal.

So, what happens with denary numbers that are greater than 9?

First convert the denary number 12 to 4-bit binary.

8	4	2	1
1	1	0	0

To convert the binary number to hexadecimal you can use simple addition again. 8 + 4 = 12. If you look at the symbol table (Table 1.1), you can see that the hexadecimal for 12 is C. Therefore, the denary number 12 is 1100 as binary and 12 as hexadecimal.

Converting the denary number to binary first isn't necessary for denary numbers that are less than 16, but it makes the process easier when you need to convert larger denary numbers.

So, how do you convert a larger denary number to hexadecimal?

CONTINUED

First, convert the denary number 201 to 8-bit binary. You know you can use 8-bit binary because the denary number is less than 256.

128	64	32	16	8	4	2	1
1	1	0	0	1	0	0	1

Each hexadecimal symbol only uses 4 bits. Therefore, you need to split the 8-bit binary number into two 4-bit binary numbers.

8	4	2	1
1	1	0	0

8	4	2	1
1	0	0	1

You then convert each 4-bit binary number again using simple addition.

8 + 4 = 12

8 + 1 = 9

The hexadecimal for 12 is C and the hexadecimal for 9 is 9. Therefore, the denary number 201 converted to binary is 11001001 and to hexadecimal is C9.

To convert a hexadecimal number to a binary number of a denary number you can just reverse the process.

To convert the hexadecimal number 5E to a binary number you need to convert each symbol to 4-bit binary. Remember E is the number 14.

8	4	2	1
0	1	0	1

5

8	4	2	1
1	1	1	0

E

Therefore, the hexadecimal 5E converted to a binary number is 01011110.

To convert the hexadecimal 5E to denary, you can use the binary number that you have just calculated.

128	64	32	16	8	4	2	1
0	1	0	1	1	1	1	0

You can add all the units together to get the denary value. 64 + 16 + 8 + 4 + 2 =v94.

The hexadecimal 5E converted to a denary number is 94.

Questions

1 Convert the hexadecimal A2 to a binary number.
2 Convert the binary number 100111010001 to hexadecimal.
3 Convert the denary number 350 to hexadecimal.
4 Convert the hexadecimal 3AC to a denary number.

Why is hexadecimal used in computer science?

Programmers often use hexadecimal when writing and analysing computer programs. They do this because it is easy for a human to read hexadecimal, rather than binary.

For example, if a programmer looked at a set of data that is stored in registers, as binary it could look like this:

10011100101110111000111011100010111101011010

The programmer may be trying to find out if the data stored matches the correct data which should be:

10011100101110111001011011100010111101011010

At a glance, it is difficult to see whether the two values match or if an error has occurred.

If that stored data is represented in hexadecimal it would be:

9CBB8EE2F5A

Comparing it to the correct value that should be:

9CBB96E2F5A

It is much easier to see that the data does not match, and the programmer can identify that an error has occurred.

The benefits of using hexadecimal rather than binary for a programmer are:

- Hexadecimal is a shorter way of representing binary, so it takes up less space on the computer screen or piece of paper.
- Hexadecimal is easier for a programmer to read, write and understand.
- Hexadecimal is easier to debug and locate errors in data.

ACTIVITY 1.1

Each computer has a **MAC address**. This MAC address is written in hexadecimal. This is one of the ways that hexadecimal is used in computer science.

Use the internet to find three more uses of hexadecimal in computer science.

KEY WORD

MAC address: a unique address that is given to a computer at the manufacturing stage that can be used to identify the computer.

1.2 Binary manipulation and negative numbers

Binary addition

You need to be able to add together two 8-bit binary numbers. When binary numbers are added together there are four rules that you will need to use:

Rule 1 is 0 + 0 = 0